Bootstrapping Endangered Language ASR with Short-Form Corpora

Christopher Bartley [1] Anton Ragni [1]

{csjbartley1, a.ragni}@sheffield.ac.uk

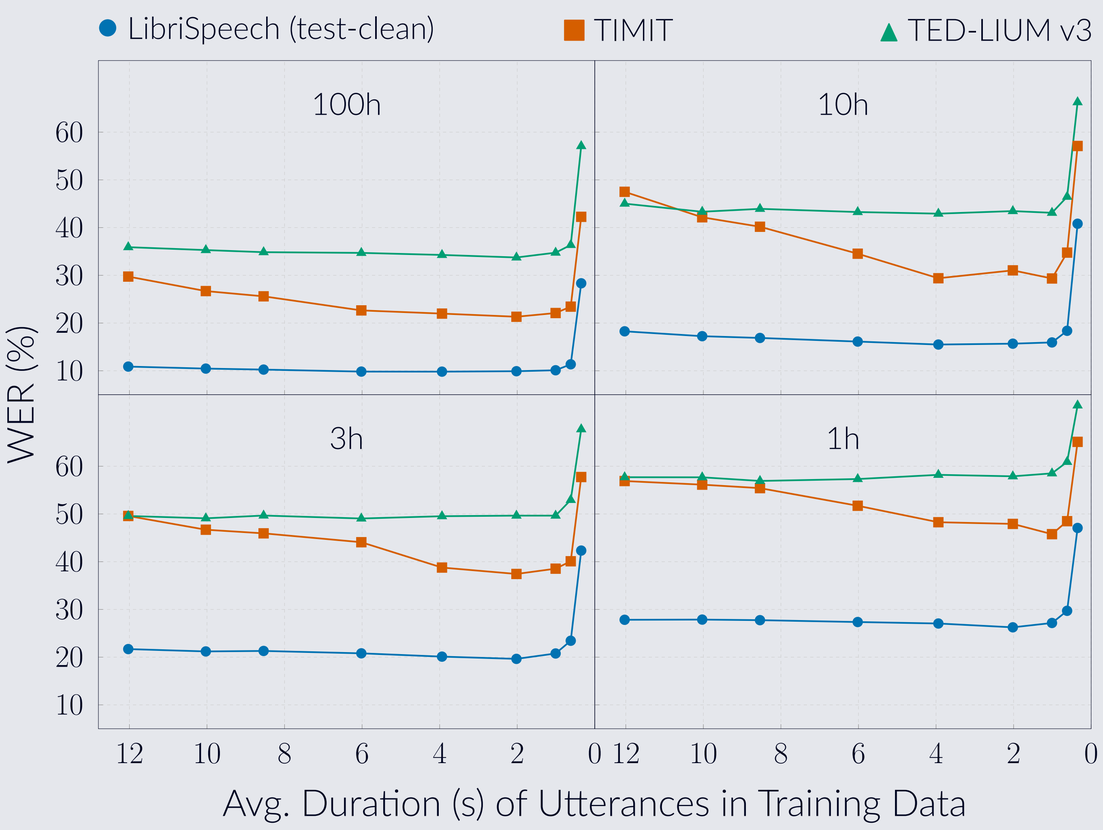
[1]University of Sheffield

Motivation

- Half of the world's 7,000 languages are endangered.
- ASR can support revival efforts [1, 2, 3].
- Most endangered languages lack utterance-level data.
- Existing transcribed resources are routinely overlooked:
  1. Short-form speech (<2 s recordings)
  2. Long-form speech (>30 s recordings)
- Short-form data is rarely used for ASR.
- Most endangered language communities lack GPU access.

Using short-form data for English ASR

Figure 1. %WERs of GMM-HMMs across LibriSpeech train-clean-100 granularities and subset sizes.



- WER stays stable down to ∼1 s segments.
- Below 1 s, WER rises sharply (avg. +20% relative).
- Effect is smaller for **OOD data** & at **lower data budgets**.
- Format becomes less important as data becomes scarce.

Do these findings generalise to endangered languages?

Endangered language data

- 5 typologically diverse endangered languages.
- Web-sourced text and speech (short and long-form).
- Test sets constructed from out-of-domain recording(s).

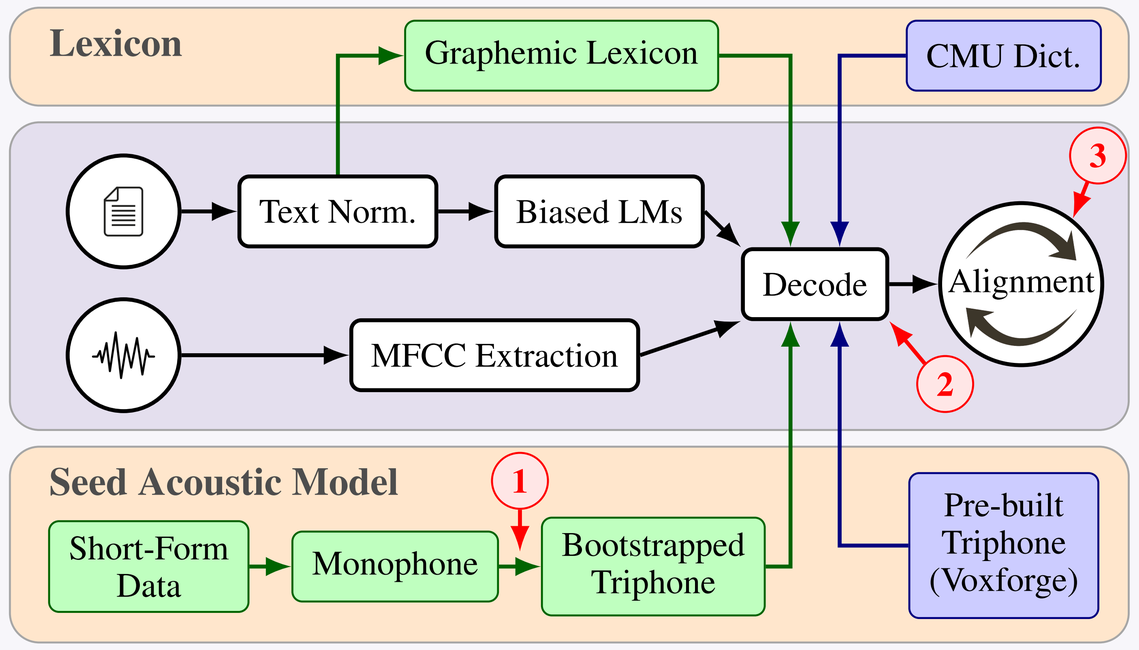
Table 1. Dataset statistics by language, data type, and quantity.

	Speech (hrs)			Text (#words)	
Language	Short	Long	Test	Train	External
Cornish	0.14	43.0	0.09	247K	2.2M
Manx	1.70	17.2	0.15	121K	5.8M
Hawaiian	0.26	59.7	0.13	139K	43.6M
Jejueo	0.11	4.0	0.02	24K	1.1M
Mohawk	2.67	2.8	0.25	15K	636K

To use the long-form data, it must first be force-aligned.

The alignment pipeline

Figure 2. Kaldi-based pipeline for force-aligning long-form transcribed speech using short-form corpora (data from Table 1, results Table 2).



Does it work?

Table 2. Alignment success rates per language (stages 1-3).

Lang.	① Mono. Ali. (%) →	② Tri. WER (%) →	③ Audio Kept (%)
Cornish	86.06 →	3.26 →	90.00
Manx	72.42 →	12.12 →	69.70
Hawaiian	85.80 →	36.75 →	23.20
Jeju	71.40 →	2.32 →	82.40
Mohawk	89.40 →	3.77 →	74.80

- Triphone achieved <5% WER for **3/5 languages**.
- Cornish retained **90% audio**; clean read-speech.
- Hawaiian retained **23% audio**; noisy broadcast data.

Can we build competitive ASR from these resources?

Our ASR vs SOTA models

Table 3. ASR %WER results on out-of-domain test sets

	SOTA Multilingual		Ours		
Language	OmniASR	Whisper	GMM	DNN+LM	Whisper-FT
Cornish	115.17	104.50	96.90	89.97	**74.23**
Manx	56.57	99.00	53.10	**24.50**	42.28
Hawaiian	36.56	67.40	26.10	**14.56**	19.47
Jeju	100.00	71.00	101.90	97.20	**68.22**
Mohawk	161.64	141.60	86.00	83.71	**68.04**

- Our GMM-HMM beats SOTA models on **4/5 languages**.
- Whisper-FT excels where **text is scarce** (Jeju, Mohawk).
- Typologically diverse languages remain challenging.

References

[1] K. Matsuura, S. Ueno, M. Mimura, S. Sakai, and T. Kawahara, "Speech corpus of ainu folklore and end-to-end speech recognition for ainu language," *CoRR*, vol. abs/2002.06675, 2020.

[2] D. Jones, "Development and evaluation of speech recognition for the Welsh language," in *Proceedings of the 4th Celtic Language Technology Workshop within LREC2022*, pp. 52–59, European Language Resources Association, 2022.

[3] V. Gupta and G. Boulianne, "Automatic transcription challenges for Inuktitut, a low-resource polysynthetic language," in *Proceedings of the Twelfth LREC*, pp. 2521–2527, European Language Resources Association, 2020.